LAB 6: Sequence Diagram – Library Management System

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Aim

To draw the sequence diagram of the Library Management System for the Return Book and Pay Fine scenario, showing how the objects interact with each other in time order.

Theory

A sequence diagram is an interaction diagram that shows how objects communicate with each other through messages, arranged in the order in which they happen. Time flows from top to bottom.

The elements used in the diagram are:

- **Actor / Object:** a participant of the interaction, drawn as a box at the top.
- **Lifeline:** the dashed vertical line below a participant that shows its existence over time.
- **Message:** a horizontal arrow from one lifeline to another, labelled with the operation that is called. A solid arrow is a call.
- **Return message:** a dashed arrow that sends the result back to the caller.
- **Combined fragment (alt):** a box that shows alternative paths. Only the path whose condition is true is executed, and the paths are separated by a dashed line.

Scenario

The scenario drawn is Return Book. The librarian enters the issue ID of the book that the member returns. The system finds the issue record and calculates the fine. If a fine is due, a fine record is created, the librarian collects the amount from the member and the fine is marked as paid. If there is no fine, nothing more is needed. In both cases the system then increases the stock of the book and confirms the return to the librarian.

Objects Used

Object	Role
Librarian	The actor who accepts the returned book.
:LibrarySystem	The control object that receives the request and coordinates the other objects.
:BookIssue	The record of the issued book. It holds the due date and calculates the fine.

:Fine	The penalty record that is created for a late return and marked as paid later.
Object	Role
:Book	The book whose available copies increase after the return.

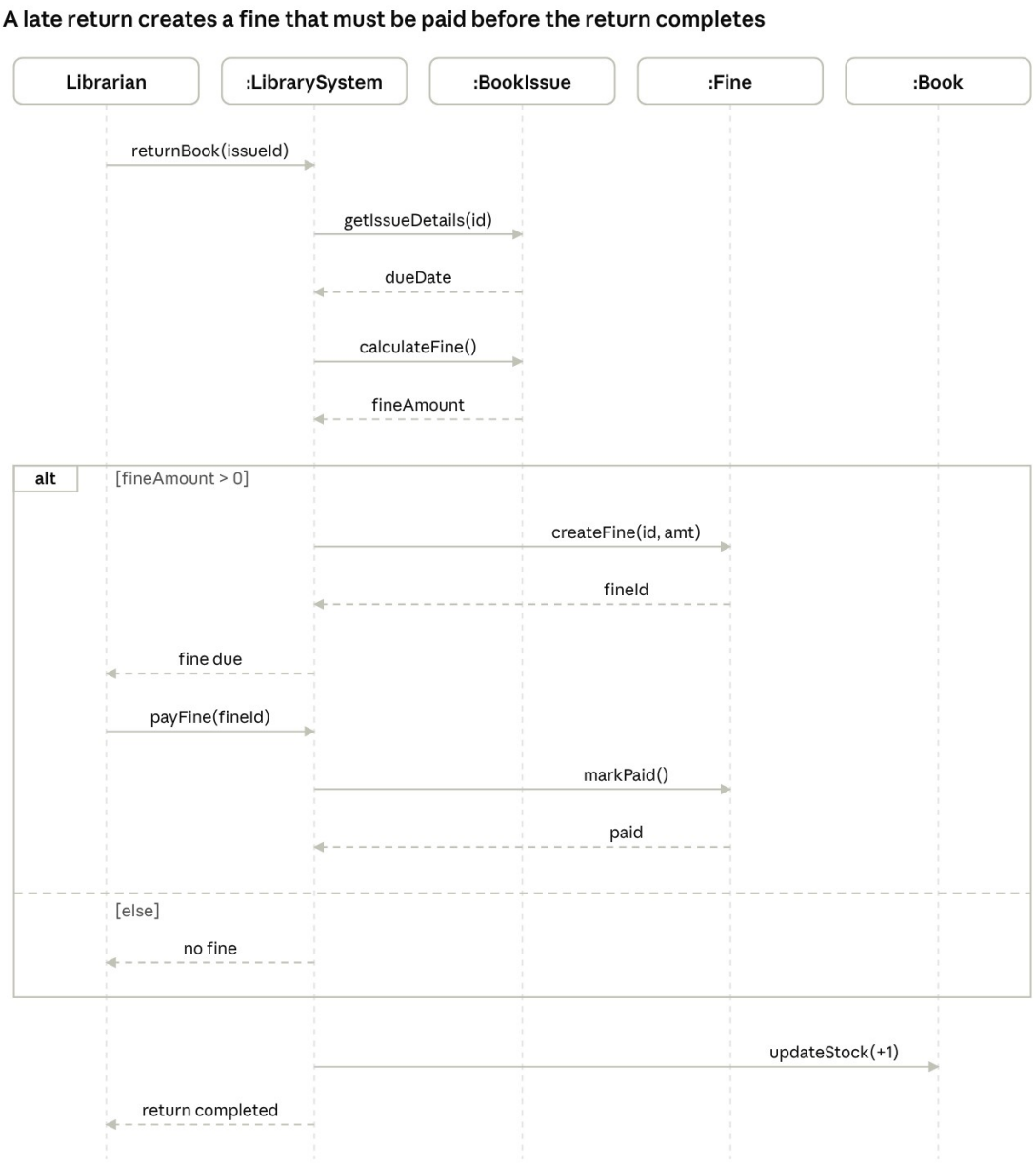
Message Sequence

No.	From	To	Message	Description
1	Librarian	:LibrarySystem	returnBook(issueId)	The librarian sends the issue ID of the returned book.
2	:LibrarySystem	:BookIssue	getIssueDetails(id)	The system asks for the details of the issue record.
3	:BookIssue	:LibrarySystem	dueDate	Return message with the due date.
4	:LibrarySystem	:BookIssue	calculateFine()	The system asks the record to calculate the fine from the due date and the return date.
5	:BookIssue	:LibrarySystem	fineAmount	Return message with the fine amount.
6	:LibrarySystem	:Fine	createFine(id, amt)	Fine due: a fine record is created.
7	:Fine	:LibrarySystem	fineId	Return message with the new fine ID.
8	:LibrarySystem	Librarian	fine due	Fine due: the librarian is told to collect the fine.
9	Librarian	:LibrarySystem	payFine(fineId)	Fine due: the librarian records the payment of the member.
10	:LibrarySystem	:Fine	markPaid()	Fine due: the fine is marked as paid.
11	:Fine	:LibrarySystem	paid	Return message that the fine is paid.
12	:LibrarySystem	Librarian	no fine	Otherwise: the librarian is told that no fine is due.

13	:LibrarySystem	:Book	updateStock(+1)	The available copies of the book are increased by one.
14	:LibrarySystem	Librarian	return completed	The completion of the return is shown to the librarian.

Sequence Diagram

The sequence diagram of the Return Book and Pay Fine scenario is shown below.



sequence diagram · 5 lifelines, 1 alt fragment

Figure 1: Sequence diagram of the Return Book and Pay Fine scenario in the Library Management

System.

Conclusion

The sequence diagram of the Return Book and Pay Fine scenario of the Library Management System was drawn. It shows the order of the messages between the Librarian, LibrarySystem, BookIssue, Fine and Book objects, and the alternative paths for a return with a fine and a return without a fine.